

AGBONLAHOR HENRY

Software & Cloud Engineer

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PROFESSIONAL SUMMARY

Backend Engineer with 5+ years of experience architecting high-concurrency, production-grade distributed systems across Fintech, E-commerce, and SaaS. Expert in **Java (Spring Boot)**, **Python (Django)**, and **Node.js (NestJS)**, with deep focus on event-driven architecture, microservices, and system observability. Proven track record reducing deployment cycles by 40%, achieving sub-second latency for real-time logistics, and building full-stack monitoring pipelines with Prometheus, Grafana, Loki, and Tempo. Skilled at mentoring cross-functional teams and leading system design for complex multi-tenant ecosystems.

CORE TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, Node.js, SQL, Bash Scripting

Frameworks: NestJS, Spring Boot, Django (DRF), FastAPI

Architecture: Microservices, Distributed Systems, Event-Driven Architecture (EDA), REST, GraphQL, gRPC, Multi-Tenant SaaS

Data & Messaging: PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis, BullMQ, Apache Kafka, RabbitMQ, Celery

Observability: Prometheus, Grafana, Sentry, Loki, Tempo, OpenTelemetry, Distributed Tracing, Log Aggregation, Monitoring

Networking: DNS, Cloudflare, CDN, VPC, Nginx, Load balancer, API Gateway, firewall

DevOps & Cloud: AWS (EC2, S3, Lambda, ECR, EKS, IAM), DigitalOcean, Docker, Kubernetes, Nginx, GitHub Actions (CI/CD), ArgoCD, Terraform

Practices: Test-Driven Development (TDD), JUnit, PyTest, Agile/Scrum, System Design, Mentorship

PROFESSIONAL EXPERIENCE

Lead Software Engineer — XstakUp Jan 2024 – current

Nigeria | Primary full-time role.

- Architected a modular multi-tenant SaaS platform supporting isolated tenant databases, RBAC authorization, audit logging, and enterprise-grade permission management.
- Designed an event-driven architecture using Kafka to decouple business workflows, enabling asynchronous processing of notifications, billing, reporting, and audit events.
- Implemented the Outbox Pattern to guarantee reliable event publication between transactional databases and Kafka, eliminating lost events during service failures.
- Built distributed tracing across all backend services using OpenTelemetry, Prometheus, Loki, Tempo, and Grafana, reducing MTTR by over 60%.
- Designed a resilient background processing infrastructure with retry policies, exponential backoff, dead-letter queues, and idempotent workers for reliable asynchronous execution.
- Introduced optimistic locking and transactional boundaries for concurrent write operations, preventing race conditions across high-concurrency workflows.

- Designed API versioning, centralized exception handling, request validation, rate limiting, and standardized error contracts for all public APIs
- Led architectural reviews for new platform capabilities, balancing scalability, maintainability, and operational cost.
- Introduced CI/CD pipelines supporting zero-downtime deployments, automated rollback, database migration validation, and feature flag rollouts.
- Mentored four backend engineers through architecture sessions, code reviews, pair programming, and engineering standards.

Lead Software Engineer (Part-Time, Concurrent) — ABX (AfrobasketXpress) Jan 2025 – April 2026

Nigeria

- Designed a complete marketplace settlement engine responsible for escrow management, vendor commissions, platform fees, refunds, and payout orchestration.
- Built an internal ledger capable of tracking every monetary movement across customer wallets, vendor balances, settlement accounts, and platform revenue.
- Implemented idempotent payout processing ensuring payment retries never generated duplicate settlements.
- Built asynchronous payout orchestration using BullMQ with retry queues, exponential backoff, dead-letter queues, and failure recovery.
- Designed event-driven financial workflows using transactional events and the Outbox Pattern to synchronize accounting operations across services.
- Engineered reconciliation jobs comparing internal ledger balances with payment gateway settlement reports.
- Implemented compensating transaction workflows for payment reversals, failed transfers, and refund processing.
- Built audit trails for every financial event including balance mutations, payout attempts, retries, and settlement confirmations.
- Optimized payout processing throughput by batching settlement operations while preserving transactional consistency.
- Orchestrated containerized deployments using Docker and Nginx on DigitalOcean, maintaining 99.9% uptime for core services.

Software Developer — DailyPayAfrica (Fintech) Jan 2023 – Dec 2023

Nigeria

- Designed and implemented a high-availability payment processing platform supporting wallet funding, transfers, merchant payments, and transaction reconciliation.
- Engineered an idempotent payment architecture preventing duplicate financial transactions across retries, network failures, and client-side request resubmissions.
- Designed a double-entry ledger system ensuring every financial movement remained balanced, traceable, and fully auditable.
- Implemented transactional boundaries using Spring Transaction Management to guarantee ACID consistency for financial operations.
- Designed automated reconciliation pipelines matching internal ledger entries against external payment gateway settlements.
- Introduced Saga-based workflows for long-running financial operations involving multiple services while maintaining eventual consistency.
- Built resilient payment retry infrastructure using exponential backoff, dead-letter queues, and compensating transactions for failed settlements.
- Reduced deployment failures by introducing feature flags, blue-green deployments, and automated rollback strategies.
- Improved payment reliability by introducing optimistic locking and distributed idempotency controls, eliminating double-spending scenarios.

Backend Developer Instructor — Univelcity Jan 2022 – Jun 2024

Nigeria

- Designed and executed a comprehensive backend curriculum, training 500+ aspiring engineers in Python and Node.js.
- Standardized DevOps training by introducing students to Docker-based workflows and GitHub Actions for automated testing.

KEY PROJECTS

Distributed Subscription Billing Platform

Lead Backend Engineer

- Architected a subscription billing engine supporting monthly, annual, metered, and usage-based pricing models.
- Designed recurring billing workflows capable of handling renewals, payment retries, subscription upgrades, downgrades, pauses, cancellations, and proration.
- Implemented idempotent billing jobs ensuring duplicate webhook deliveries never produced duplicate invoices or charges.
- Built webhook processing infrastructure with signature verification, retry handling, deduplication, and event ordering guarantees.
- Implemented invoice generation, tax calculation, coupon support, credits, and subscription lifecycle management.
- Designed dunning management workflows including retry schedules, customer notifications, grace periods, and automated subscription suspension.
- Built event-driven billing architecture integrating payment gateways through asynchronous domain events.
- Introduced Outbox Pattern and Inbox Pattern to guarantee exactly-once event processing across billing services.
- Achieved high availability through horizontally scalable workers, queue partitioning, and distributed scheduling.

Financial Ledger Engine

Lead Backend Engineer · Xstakup ·

- Designed an immutable double-entry accounting ledger where every financial operation balanced debits and credits.
- Implemented append-only ledger entries with immutable audit history.
- Built reconciliation pipelines capable of detecting balance drift.
- Designed journal entries supporting transfers, reversals, adjustments, settlements, and chargebacks.
- Introduced balance snapshots and aggregation strategies reducing account balance lookup latency by over 80%.

Real-Time Logistics & Navigation Engine

Lead Role · ABX (AfrobasketXpress) · Part-Time Concurrent Engagement

- Architected a high-frequency location tracking system using MQTT (Mosquitto) and WebSockets to achieve sub-second latency for delivery partner updates.
- Implemented distance calculation and route optimization using Google Directions and Navigation APIs for precise ETA predictions.
- Engineered a throttled ingestion layer to handle high-frequency coordinate bursts, preventing backend saturation and ensuring 99.9% message durability during peak delivery windows.
- Developed a custom React Native/Expo hook for animated map markers, synchronizing backend GPS streams with smooth UI transitions.

Automated Vendor Payout & Reconciliation System

Lead Role · ABX (AfrobasketXpress) · Part-Time Concurrent Engagement

- Designed a secure financial workflow for multi-vendor settlements using NestJS event listeners and Django signals to trigger automated reconciliation upon order completion.
- Eliminated 95% of manual accounting overhead by implementing a ledger-based fund reconciliation system with idempotent transaction processing.
- Integrated BullMQ for reliable background processing of payout batches, ensuring failover recovery and transaction integrity.

EDUCATION

Bachelor of Engineering (B.Eng), Mechanical Engineering

Ambrose Alli University, Ekpoma, Nigeria — 2019

Cloud Developer Nanodegree

Udacity — 2022

ADDITIONAL HIGHLIGHTS

- Strong ownership mindset with the ability to deliver features from design to deployment.
- Comfortable working in high-focus, high-output teams with minimal supervision.
- Experience building systems at fintech scale with production Kafka usage and full-stack observability tooling.